From the Hamiltonian to the observables: the formulas behind every number

fourterminal: derivations, sources and code locations

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This document covers every quantity the Scharf-model studies compute. For each one it gives the equation, a derivation, the physical reasoning, a literature source where one exists, and the code that implements it. In this PDF a code link opens the file at that line in VS Code. A section sign (§) links to that section. The PDF is built from DERIVATIONS.md by derivations_pdf.py; the Markdown is the source.

What the source tags mean

- [**source, verified**]: the formula is in the cited paper. I checked it against the paper's own LaTeX source from arXiv; equation numbers were checked against the printed PDF.
- [**derived here**]: I derived it myself, from the cited equations or from scratch. These derivations are not in the paper they build on.
- [**model choice**]: a decision made in this code, not a physical law.

Everything is in the lattice units of `my_functions` unless stated otherwise (energies in units of the hopping t, lengths in units of a). The conversion to meV and nm is §1.3.

0. The pipeline at a glance

```
PhysParams (meV, nm, T)          ldos.py          §1.3  units, E_Z <-> B, E_T, field angle
        |  .params() / .params_transport()
        v
Params (lattice units)  ->  4x4 BdG blocks       hamiltonians.py   §1    H = onsite + hoppings
        |
        +--> TRANSPORT GEOMETRY (finite centre, semi-infinite probes and SC ribbons)
        |      leads.py   §2.1  surface Green's functions (Sancho-Rubio), self-energies, Gamma
        |      rgf.py     §2.2  G of the centre by RGF + one exact Dyson step for the ribbons
        |      rgf.py     §3    channels T_ee, T_he, ... (Caroli formula)
        |      tools.py   §4    kappa, G, local Andreev, det r      transport.py §4.4  finite T
        |      study_*.py §5    two-Majorana model: x = eps/Gamma, Gamma_eff, p, plateau criterion
        |
        +--> CROSS-SECTION (infinitely long junction, Bloch k along it)
               tools.py           §6    H(k), gaps, Pfaffian invariant Q
               study_oscillation  §7    complex k (xi_M, k_osc), gap dip, closed box eps,
                                        overlap law, end modes (study_mixed [amplitude])
```

The two branches meet in §5. The closed-box splitting ε from §7, set against the transport output of §4, tests the two-Majorana picture without any free parameter.

1. The model

1.1 Continuum Hamiltonian

[**source, verified**] Scharf et al., PRB 99, 214503 (2019), arXiv:1904.08981, **Eq. (1)**. The basis is $\Psi = (\psi_\uparrow, \psi_\downarrow, \psi_\downarrow^\dagger, -\psi_\uparrow^\dagger)$, the same order the code uses:

$$H_{\mathrm{BdG}} = \Big[\tfrac{p_x^2+p_y^2}{2m} + \tfrac{\alpha}{\hbar}(s_y p_x - s_x p_y) + \tfrac{\beta}{\hbar}(s_x p_x - s_y p_y) + \tfrac{m(\alpha^2+\beta^2)}{2\hbar^2} - \mu_S\Big]\tau_z + (V_0\tau_z - \mathbf{E}_Z\cdot\mathbf{s})h(x) + \Delta(x)\big[\tau_x\cos\Phi(x) - \tau_y\sin\Phi(x)\big].$$

The paper's x is across the junction (width W) and its y is along it (length L). $h(x) = 1$ in the normal (N) strip, and $\Delta(x) = \Delta$ only in the superconducting (S) regions. The paper writes the N potential as $V_0 = \mu_S - \mu_N$; the code instead gives each region its own μ.

Physics.

- In this basis, kinetic energy, chemical potential and SOC are odd under particle–hole conjugation ($\propto \tau_z$), while the Zeeman term is even ($\propto \tau_0$).
- An in-plane field together with a phase difference near π drives the quasi-1D N strip into class D topological superconductivity, with one Majorana at each end of the strip. Scharf, Sec.

III: "As has been demonstrated in Ref. [Pientka et al., PRX 7, 021032 (2017)] … an in-plane Zeeman term … can induce topological superconductivity", in a region around $\phi = \pi$.

The axis swap. `my_functions` counts its x along the interfaces. So package x = paper y and package y = paper x; the swap is the two lines ldos.py:182-183. **[derived here]** Substituting $p_x^{\text{paper}} = p_y$ and $p_y^{\text{paper}} = p_x$ into Eq. (1) gives the SOC in package coordinates ('100' setup):

$$H_{\text{soc}} = -(\alpha s_x + \beta s_y)\, k_x + (\beta s_x + \alpha s_y)\, k_y,$$

which is the form stated in ldos.py:45-46 and built in hamiltonians.py:79-80 and :93-94.

1.2 Lattice discretisation

[derived here] This is the standard finite-difference lattice; the continuum limit below is the proof that it reproduces Eq. (1).

Onsite block hamiltonians.py:25-47:

$$h_{\text{on}} = [4t - \mu + \tfrac{\alpha^2+\beta^2}{4}]\tau_z + E_Z(\cos\theta\, s_x - \sin\theta\, s_y)\,\tau_0 + \Delta\,(e^{i\phi}\tau_+ + \text{h.c.}).$$

Here α, β are in lattice units. The pairing sits on the elements (0,2) and (1,3): $\Delta e^{i\phi}$ in the basis above. Scharf's $\Delta[\tau_x \cos\Phi - \tau_y \sin\Phi]$ is exactly $\begin{pmatrix} 0 & \Delta e^{i\Phi} \\ \Delta e^{-i\Phi} & 0 \end{pmatrix}$ in τ space.

Hopping to the $+x$ / $+y$ neighbour hamiltonians.py:54-64:

$$V = -t\,\tau_z + M \oplus (-M), \qquad M = -\tfrac{i}{2}\,(c_x s_x + c_y s_y).$$

The SOC block appears as $+M$ in the electron sector and $-M$ in the hole sector.

Continuum limit. With $H(\mathbf{k}) = h_{\text{on}} + \sum_d (V_d e^{ik_d} + V_d^\dagger e^{-ik_d})$:

- kinetic part: $4t - 2t\cos k_x - 2t\cos k_y \simeq t(k_x^2 + k_y^2)$, i.e. $t\,a^2k^2 = \hbar^2k^2/2m$ for $t = \hbar^2/2ma^2$;
- SOC part: $Me^{ik} + M^\dagger e^{-ik} = \sin k\,(c_x s_x + c_y s_y) \simeq k\,(\mathbf{c}\cdot\mathbf{s})$, which is Eq. (1) after the axis swap;
- hole sector: $-s_y h^*(-\mathbf{k}) s_y = -h_{\text{soc}}(\mathbf{k})$ for a time-reversal-even SOC term, hence the $-M$;
- the constant $(\alpha^2+\beta^2)/4$ in lattice units equals $(\alpha^2+\beta^2)/(4a^2t) = m(\alpha^2+\beta^2)/2\hbar^2$ in physical units. That is Scharf's SOC shift, so μ is measured from the bottom of the lower SOC band (CLAUDE.md, "constant SOC shift").

Interfaces [model choice] hamiltonians.py:191-210. A bond of strength t_b between regions (probe bond τ = `tc_barr_transport`, S/N bond t_c) is:

- for $t_b \le 1$: $t_b \times [-\tau_z + \text{SOC}]$ (a barrier weakens the whole hop, spin rotation included);
- for $t_b > 1$: $-t_b\tau_z + \text{SOC}$.

At $t_b = 1$ it is an ordinary bulk bond. Why this matters: CLAUDE.md, "INTERFACE BONDS SCALE AS A WHOLE".

Assembly.

- Central region: junction.py:29-64. Probes: one column of n_y sites per cell; SC ribbons: one row of n_x sites per cell.
- Phase: ϕ sits on the top ribbon, 0 on the bottom one (junction.py:56-57). That is a gauge-equivalent choice to Scharf's symmetric $\Phi(x)$.
- Zeeman field: only in the N strip (`Bxy_c`, ldos.py:218), as $h(x)$ in Eq. (1).
- Disorder (optional): $V_i\tau_z$ per site, junction.py:81-102.

1.3 Units and derived scales

Code: ldos.py:142-231.

quantity	formula	source	code
hopping	$t = \hbar^2/2ma^2$ (2.507 meV at $a = 20$ nm, $m = 0.038m_0$)	definition (§1.2)	ldos.py:143
lattice parameters	$\mu_t = \mu/t$, $\Delta_t = \Delta/t$, $\alpha_t = \alpha/(at)$	follows from §1.2	ldos.py:208-221
Zeeman energy	$E_Z = g\mu_B B/2$	**Scharf, Sec. VII** ("$\mathbf{E}_Z = g\mu_B\mathbf{B}/2$")	ldos.py:148-154
Thouless energy	$E_T = \frac{\pi}{2}\frac{\hbar v_F}{W}$, v_F of the N strip	**Scharf, Eq. (8)** and the text after it	ldos.py:157-159
field angle	$\mathbf{E}_Z \perp \mathbf{n}_{\rm soc}$: $\theta_z = \arctan 2(\alpha, \beta)$ ('100')	**[derived here]**, below	ldos.py:161-165
mixed SOC	$(\alpha, \beta) =$ 16 $(\cos 0.15\pi,\ \sin 0.15\pi)$ meV nm	[model choice] (the thesis case)	tools.py:72-74
temperature	1 μeV/k_B = 11.6 mK	$k_B = 86.17$ μeV/K	tools.py:355

Field angle [derived here].

- For propagation along the junction (k_x), the SOC field is $\mathbf{n}_{\rm soc} \propto (\alpha, \beta)$ (§1.1).
- The code's Zeeman term points along $(\cos\theta, -\sin\theta)$.
- Perpendicular means $\alpha\cos\theta - \beta\sin\theta = 0$, i.e. $\theta = \arctan 2(\alpha, \beta)$.

The extra $+\pi$ in `case()` (tools.py:68) reverses the field. That mirrors the maps about $\phi = \pi$ and compensates the top/bottom orientation (CLAUDE.md, "MATCH TO THE PAPER").

Thouless energy in lattice units [derived here]. $\hbar v_F = \hbar\sqrt{2\mu/m} = 2a\sqrt{\mu t}$ and $W = n_{\rm across}\, a$, so $E_T = \frac{\pi}{2}\cdot 2\sqrt{\mu t}/n_{\rm across}$, which is the code line.

2. Green's functions of the transport geometry

2.1 Semi-infinite leads: surface Green's function, self-energy, coupling

Equations. A lead is a chain of identical cells: onsite block H_0, outward hop V. Its surface Green's function $g_s = (z - \varepsilon_s)^{-1}$ with $z = E + i\eta$ satisfies the self-consistency (Dyson) equation

$$\varepsilon_s = H_0 + V\,(z - \varepsilon_s)^{-1} V^\dagger.$$

[derived here] This holds because removing the surface cell leaves an identical semi-infinite lead. The coupling to the centre (hop V_c) gives the self-energy and the coupling matrix:

$$\Sigma(z) = V_c^\dagger\, g_s(z)\, V_c, \qquad \Gamma = i(\Sigma - \Sigma^\dagger).$$

[source, verified] Lewenkopf & Mucciolo, J. Comput. Electron. 12, 203 (2013), arXiv:1304.3934: **Eq. (16)** gives $\Gamma_{L,R} = i[\Sigma^r - \Sigma^{r\dagger}]$, **Eq. (17)** gives $\Sigma^r_L = u_L g^r_L u_L^\dagger$, and **Eq. (18)** defines the surface Green's function.

Algorithm. Sancho–Rubio decimation: M. P. López Sancho, J. M. López Sancho and J. Rubio, J. Phys. F 15, 851 (1985). I did not check that paper's text; it is cited for exactly this method by Lewenkopf–Mucciolo (section on the decimation method) and by Gresta et al. (ref. 61). Each iteration eliminates every second cell and renormalises

$$\varepsilon_s \leftarrow \varepsilon_s + \alpha g \beta, \qquad \varepsilon_b \leftarrow \varepsilon_b + \alpha g \beta + \beta g \alpha, \qquad \alpha \leftarrow \alpha g \alpha, \quad \beta \leftarrow \beta g \beta, \qquad g = (z - \varepsilon_b)^{-1},$$

starting from $\alpha = V$, $\beta = V^\dagger$. The effective chain length doubles at every step.

Physics.

- Σ is what the infinite lead does to the centre. Its Hermitian part shifts levels; its anti-Hermitian part Γ is the escape rate into the lead, the broadening of a level coupled to it.
- The small η acts as a uniform absorption. It must stay far below every physical width; CLAUDE.md ("Numerics") has the measured consequences.

Code

- decimation: leads.py:48-80;
- residual check of the Dyson equation above: leads.py:83-87;
- Newton refinement: leads.py:90-120;
- eigenmode fallback: leads.py:123-135;
- failed solves become NaN and are counted in `N_UNRELIABLE`: leads.py:174-179;
- Σ: leads.py:239-242;
- Γ: rgf.py:293-294.

2.2 Green's function of the centre: RGF plus one exact Dyson step

Equation. The quantity needed is

$$G^r = \left(z - H_C - \Sigma_L - \Sigma_R - \Sigma_T - \Sigma_B\right)^{-1},$$

but only on the probe columns $(G_{LL}, G_{LR}, G_{RL}, G_{RR})$.

Step 1: RGF along the junction. $A_0 = z - H_C - \Sigma_L - \Sigma_R$ is block-tridiagonal in the columns, so $g_0 = A_0^{-1}$ follows from left and right sweeps:

$$g_i^L = \left(A_{ii} - V^\dagger g_{i-1}^L V\right)^{-1}, \qquad g_i^R = \left(A_{ii} - V g_{i+1}^R V^\dagger\right)^{-1}.$$

The off-diagonal blocks follow from $G_{ij} = g_i^R V^\dagger G_{i-1,j}$ for $i > j$ and $G_{ij} = g_i^L V G_{i+1,j}$ for $i < j$.

- **[source, verified]** Lewenkopf–Mucciolo, Sec. 3: built from the Dyson equations $G = G^{(0)} + G^{(0)}VG$, their Eqs. (15)–(16) in their slice notation.
- **Code:** rgf.py:115-169; the recursions are :138-144.

Step 2: the two SC ribbons in one Dyson step. [derived here] This is exact algebra: the matrix inversion lemma applied to $\Sigma_{T,B} = W^\dagger g_s W$, which lives on the boundary rows P.

$$G_{QQ} = g_{0,QQ} + (g_{0,QP}W^\dagger)\left[g_s^{-1} - W g_{0,PP} W^\dagger\right]^{-1}(W g_{0,PQ}), \quad Q = \text{probe columns},\ P = \text{ribbon rows}.$$

- It uses $g_s^{-1} = z - \varepsilon_s$, which stays finite even when the ribbon has a Majorana pole at $E = 0$. That is why it is stable (docstring, rgf.py:25-34).
- **Code:** rgf.py:342-350; its pieces are built in :317-319.

The phase costs nothing [derived here]. ϕ enters only the top ribbon's pairing. A gauge transformation $U = \mathrm{diag}(1, 1, e^{i\delta\phi}, e^{i\delta\phi})$ per site maps the ribbon at ϕ_{ref} to the ribbon at ϕ, and U commutes with the normal-state coupling W, so

$$g_T^{-1}(\phi) = U^\dagger g_T^{-1}(\phi_{\mathrm{ref}})\, U.$$

The ribbon decimation therefore runs once; see rgf.py:337-340.

3. Transmission channels (Caroli formula in Nambu space)

[**source, verified**] Gresta et al., PRB 114, 125405 (2026), arXiv:2602.12947, **Eq. (7)**:

$$\mathcal{T}^{\alpha\beta}_{RL}(\varepsilon) = \mathrm{Tr}[\Gamma_R\, \mathcal{G}^{r;\alpha\beta}_{RL}\, \Gamma_L\, \mathcal{G}^{a;\beta\alpha}_{LR}],$$

"associated with a quasiparticle scattering from type $\beta \in \{e, h\}$ from L … to a type α in R". The trace runs over spin and position. This is the Caroli formula (Caroli et al., J. Phys. C 4, 916 (1971)); for the normal-state version see also Lewenkopf–Mucciolo, **Eq. (15)**.

Physics. Γ is block-diagonal in electron/hole because the probes are normal metals. Projecting both Γ's and G onto the e or h sectors splits the total transmission into processes:

key in the code	process	physical meaning
`ee`, `hh`	$e \to e$, $h \to h$ across	elastic cotunnelling (EC)
`he_cross`, `eh_cross`	$e \to h$, $h \to e$ across	crossed Andreev reflection (CAR): a Cooper pair is split between the two probes
`he_local`, `eh_local`	$e \to h$ in the same probe	local Andreev reflection (LAR)

Particle–hole symmetry. [**source, verified**] Klees et al., PRB 110, 064517 (2024), arXiv:2306.17845, Sec. III: "$\mathcal{T}^{\tau_1\tau_2}_{\ell_1\ell_2}(\varepsilon) = \mathcal{T}^{\bar\tau_1\bar\tau_2}_{\ell_1\ell_2}(-\varepsilon)$". At $E = 0$ this forces `ee` = `hh` and `he_cross` = `eh_cross` exactly. The code uses this as a regression check (transport.py:167-191); it is not a breakdown detector.

Code

- RGF solver: rgf.py:352-386; the six formulas are :373-380;
- dense reference implementation, for comparison: junction.py:215-234.

Direction. The studies call `channels_at_phi(phi, 'left')`, i.e. transmission from R into L.

- [**derived here**] For the heat signal the direction does not matter at $E = 0$ as long as only the two probes carry open channels (the trivial ribbons are gapped). Unitarity of the L/R block of S gives $\mathrm{Tr}\, tt^\dagger = \mathrm{Tr}\, t't'^\dagger$, and particle–hole symmetry makes each trace equal to $2(\mathcal{T}^{ee} + \mathcal{T}^{he})$.

4. Observables at the probes

4.1 Nonlocal charge conductance G

[**source, verified**] Gresta et al., **Eqs. (3) and (5)**:

$$G = G_0 \int d\varepsilon \Big(-\frac{\partial f}{\partial \varepsilon} \Big) \mathcal{T}^E, \qquad \mathcal{T}^E = \mathcal{T}^{ee} - \mathcal{T}^{he}, \qquad G_0 = e^2/h.$$

At $T \to 0$ this gives $G/G_0 = \mathcal{T}^{ee} - \mathcal{T}^{he}$ at $E = 0$.

Sign [derived here]. The charge current of Klees et al., App. C (the final I_R, J_R formula, after their particle–hole simplification) is

$$I_R = \frac{e}{h} \int d\varepsilon\, [\mathcal{T}^{ee}_{RL}(f^e_R - f^e_L) + \mathcal{T}^{eh}_{RL}(f^e_R - f^h_L) + \mathcal{T}^{eh}_{RR}(f^e_R - f^h_R)].$$

Differentiating with respect to V_L at $T = 0$ gives $\partial I_R/\partial V_L = -G_0(\mathcal{T}^{ee} - \mathcal{T}^{he})(0)$. So the code's `G` $= \mathcal{T}^{ee} - \mathcal{T}^{he}$ is Gresta's G, which is minus the nonlocal dI/dV in units of e^2/h.

Physics. EC carries charge $+e$ across, CAR carries $-e$. A Majorana pair transfers an electron as an electron or as a hole with **equal** probability. Nilsson et al.: "with equal probabilities p", see §5.1. So $G = 0$ exactly. Charge neutrality is the Majorana fingerprint that the code tests as $|G| < 0.1\,\kappa$.

Code: `G = ee - he_cross`, tools.py:290. The finite-bias version is `current()` transport.py:52-66, with the $\frac{1}{2}$ that removes the Nambu double counting.

4.2 Thermal conductance κ

[**source, verified**] Gresta et al., **Eqs. (4) and (6)**:

$$\mathcal{T}^Q = \mathcal{T}^{ee} + \mathcal{T}^{he}, \qquad \kappa_0 = \frac{\pi^2 k_B^2 T}{3h},$$

with the stated $T \to 0$ limit $\kappa/\kappa_0 = \mathcal{T}^Q$.

[**source, verified**] Klees et al., Sec. IV: the heat transport coefficient uses $\mathcal{T}_{22} = \mathcal{T}^{ee}_{RL} + \mathcal{T}^{eh}_{RL}$. In their words, "a temperature gradient does not generate LAR contributions to the thermal current J_R".

Derivation [derived here] (from the J_R line of Klees et al., App. C):

1. Take a thermal bias only: $\mu_L = \mu_R = 0$, so for each probe $f^e = f^h = f_T$.
2. The LAR term $\propto f^e_R - f^h_R$ then vanishes identically, and

$$J_R = \frac{1}{h}\int d\varepsilon\, \varepsilon\, [\mathcal{T}^{ee}_{RL} + \mathcal{T}^{eh}_{RL}](f_R - f_L).$$

3. Use $\partial f/\partial T = (\varepsilon/T)(-\partial f/\partial \varepsilon)$ and $x = \varepsilon/k_B T$:

$$\kappa = \frac{k_B^2 T}{h}\int d\varepsilon\, x^2\Big(-\frac{\partial f}{\partial \varepsilon}\Big)\mathcal{T}^Q \quad \Rightarrow \quad \frac{\kappa}{\kappa_0} = \int dx\, \frac{3}{\pi^2}\,\frac{x^2}{4\cosh^2(x/2)}\,\mathcal{T}^Q(xk_BT).$$

4. The kernel integrates to one, because $\int x^2/(4\cosh^2(x/2))\,dx = \pi^2/3$. So $T \to 0$ gives $\kappa/\kappa_0 = \mathcal{T}^Q(0)$.
5. $\mathcal{T}^{eh}_{RL}$ (Klees) and $\mathcal{T}^{he}_{RL}$ (Gresta) differ only by $\varepsilon \to -\varepsilon$ (particle–hole symmetry). They are equal at $E = 0$ and give the same integral with the even kernel.

   Note on Gresta's Eq. (4) as printed. It reads $\kappa = \kappa_0 \int (\varepsilon/k_BT)^2(-\partial f/\partial\varepsilon)\mathcal{T}^Q$. Since the kernel integrates to $\pi^2/3$, that would give $\kappa/\kappa_0 \to \frac{\pi^2}{3}\mathcal{T}^Q$ at $T \to 0$, contradicting their own stated limit. The prefactor should be $k_B^2 T/h = 3\kappa_0/\pi^2$. The code uses the normalised kernel above, consistent with the stated limit.

This κ is taken at zero voltage. The open-circuit value adds a thermoelectric correction ($K = 3K_0 \det\mathcal{L}/(\pi^2\mathcal{L}_{11})$ in Klees et al.), which Gresta neglects and so does the code.

Physics.

- A spin-degenerate normal channel with $T = 1$ gives $\mathcal{T}^{ee} = 2$, so $\kappa = 2\kappa_0$.
- One Majorana carries half a fermionic degree of freedom. Gresta et al., Sec. I: "A single Majorana bound state carries only half of the degrees of freedom of a conventional fermionic mode … the thermal conductance reaches the universal quantized value, $\kappa = 0.5\kappa_0$". This is the chiral (Gresta) case.
- In the localized-pair (Scharf) case, ½ is reached only on resonance (§5.1).

Code: `kappa = ee + he_cross`, tools.py:290; `linear_response` compute.py:35; finite-T kernels in §4.4.

4.3 Local Andreev conductance

[**derived here**] from the LAR term of `current()`, transport.py:63:

- The current is $I^{\mathrm{LAR}} = \frac{1}{2}\int (R_{eh} + R_{he})(f_e - f_h)$, and $\partial(f_e - f_h)/\partial V = 2(-\partial f/\partial E)$.
- Hence $G_{\mathrm{LAR}} = G_0 \int (R_{eh} + R_{he})(-\partial f/\partial E) \to G_0\,(R_{eh} + R_{he})(0) = 2G_0 R_{he}(0)$.

Physics. An isolated Majorana gives perfect Andreev reflection, $R_{he}(0) = 1$, hence $2e^2/h$.

- **[source, verified]** Law, Lee & Ng, PRL 103, 237001 (2009), arXiv:0907.1909: "the presence of a vortex in the superconductor changes the conductance to $2e^2/h$" (resonant Andreev reflection via a Majorana).
- This quantity needs only one Majorana, so it is not sensitive to the splitting. Its plateau is real but not specific to Majoranas (CLAUDE.md, "THE PLATEAU THAT DOES EXIST").

Code: `andreev = eh_local + he_local`, tools.py:290; finite T in tools.py:376-385.

4.4 Finite temperature

Equations: the kernels of §4.1 and §4.2,

$$W_{\mathrm{el}}(x) = \frac{1}{4\cosh^2(x/2)}, \qquad W_{\mathrm{th}}(x) = \frac{3}{\pi^2}x^2 W_{\mathrm{el}}(x).$$

They are evaluated with the trapezoid rule in x, and $E<0$ is folded onto $E>0$ by particle–hole symmetry. That is why `thermal()` sums `ee+he+hh+eh` over $E \geq 0$, with weight ½ at $T=0$.

- **Code:** transport.py:118-164 and tools.py:370-391.
- Grid rule: the energy grid must reach about $16\,k_B T$.

4.5 $\det r$: the topological number seen by a probe

Equations.

- **[source, verified]** Lewenkopf–Mucciolo, **Eqs. (5) and (8)** (Fisher–Lee on a lattice): $S_{ab} = -\delta_{ab} + i\sqrt{\gamma_a\gamma_b}\,\langle\chi_a|G^r|\chi_b\rangle$, with $\Gamma = \sum_a |\chi_a\rangle\gamma_a\langle\chi_a|$.
- In the eigenbasis of Γ_L, restricted to its open channels, the left-probe reflection matrix is therefore

$$r = -1 + i\,\Gamma_L^{1/2}\,G_{LL}\,\Gamma_L^{1/2}.$$

- **[source, verified]** Fulga, Hassler, Akhmerov & Beenakker, PRB 83, 155429 (2011), arXiv:1101.1749, **Table I**: for class D the topological quantum number is $\mathcal{Q} = \operatorname{sign}\det r$, with r real in the Majorana basis. Sec. III: "The determinant is unchanged by the change of basis". That is why the code may take $\det r$ in the Γ eigenbasis, and why the result is real up to rounding.

Physics.

- $\det r = -1$: the probe sees one isolated Majorana (perfect Andreev reflection, §4.3).
- $\det r = +1$: it sees none, or a hybridised pair.
- **[derived here]** $\det r = 0$ means r is singular, so one eigenchannel is fully transmitted. Then $\kappa = \frac{1}{2}\sum_n T_n \geq \frac{1}{2}$ (the trace over e and h channels counts each process twice).
- In a finite junction $\det r$ follows the Majoranas' overlap, not only the bulk phase (§5.1).

Code: tools.py:266-277.

5. The two-Majorana model and what is extracted from it

5.1 Scattering through two coupled Majoranas

[source, verified] Nilsson, Akhmerov & Beenakker, PRL 101, 120403 (2008), arXiv:0806.4889:

- **Eq. (1):** $H_M = iE_M\gamma_1\gamma_2$, i.e. a splitting $\pm E_M$.
- **Eq. (2)** (Mahaux–Weidenmüller): $S(E) = 1 + 2\pi i W^\dagger (H_M - E - i\pi W W^\dagger)^{-1} W$.
- **Eq. (3):** lead i couples only to γ_i, with equal amplitude to electrons and holes.
- **Eqs. (4)–(6):**

$$S = \begin{pmatrix} 1+A & A \\ A & 1+A \end{pmatrix}, \qquad A = \frac{1}{Z}\begin{pmatrix} i\Gamma_1(E+i\Gamma_2) & -E_M\sqrt{\Gamma_1\Gamma_2} \\ E_M\sqrt{\Gamma_1\Gamma_2} & i\Gamma_2(E+i\Gamma_1) \end{pmatrix},$$
$$Z = E_M^2 - (E+i\Gamma_1)(E+i\Gamma_2), \qquad \Gamma_i = 2\pi w_i^2.$$

The general form of Eq. (2) and the particle–hole constraint $S(0)$ real orthogonal for $\mathcal{C}^2 = +1$ are also in Beenakker, RMP 87, 1037 (2015), arXiv:1407.2131, section on the scattering matrix (Cayley transform).

At $E = 0$ [derived here], writing $x^2 = E_M^2/(\Gamma_1\Gamma_2)$, so $Z = E_M^2 + \Gamma_1\Gamma_2$:

$$\mathcal{T}^{ee} = \mathcal{T}^{he} = |A_{21}|^2 = \frac{x^2}{(1+x^2)^2} \;\Rightarrow\; \boxed{\kappa/\kappa_0 = \frac{2x^2}{(1+x^2)^2}, \quad G = 0}$$

$$A_{11} = -\frac{1}{1+x^2} \;\Rightarrow\; \det r = (1+A_{11})^2 - A_{11}^2 = 1 + 2A_{11} = \boxed{\frac{x^2-1}{x^2+1}}$$

$$\Rightarrow\; \boxed{\kappa = \tfrac{1}{2}(1 - \det r^2)} \qquad \text{and} \qquad x = \sqrt{\frac{1+\det r}{1-\det r}}.$$

Unitarity check: $R_{ee} + R_{he} + \mathcal{T}^{ee} + \mathcal{T}^{he} = [x^4 + 1 + 2x^2]/(1+x^2)^2 = 1$.

Physics.

- κ reaches ½ **only** at $x = 1$ ($\varepsilon = \Gamma$) and is smaller on both sides.
- $\varepsilon \to 0$ gives $\kappa \to 0$: the two Majoranas decouple and no heat crosses.
- $\varepsilon \gg \Gamma$ gives $\kappa \to 0$: the pair is an ordinary off-resonant level.
- So $\kappa = \frac{1}{2}$ is a **resonance in** ε/Γ, not a protected plateau (CLAUDE.md, "resonance, not a topological plateau").

Finite energy [derived here] (used by the fingerprint finite-energy cell). For $\Gamma_1 = \Gamma_2 = \Gamma$:

- $\kappa(E) = 2E_M^2\Gamma^2/|Z|^2$ with $|Z|^2 = (E_M^2 - E^2 + \Gamma^2)^2 + 4E^2\Gamma^2$.
- Minimising over $u = E^2$ gives $E^* = \sqrt{E_M^2 - \Gamma^2}$ and $|Z|^2_{\min} = 4E_M^2\Gamma^2$.
- So $\max_E \kappa(E) = \frac{1}{2}$ for every $E_M \geq \Gamma$.
- Code: `resonance_point`, study_oscillation.py:476.

Code that uses these relations

- x from $\det r$: study_tuning.py:458-467 (`x_of`) and study_mixed.py:151;
- $\kappa(x)$: study_oscillation.py:453 (`kappa_model`) and the collapse figure slides_nov.py:417.

5.2 Measuring Γ directly: the Andreev Lorentzian

[derived here] from Nilsson Eqs. (4)–(6) with $E_M = 0$:

$$A_{11} = \frac{-i\Gamma_1}{E + i\Gamma_1} \;\Rightarrow\; R_{he}(E) = |A_{11}|^2 = \frac{\Gamma_1^2}{E^2 + \Gamma_1^2}.$$

So Γ is the **half width at half maximum** of the local Andreev signal at an isolated Majorana (at an ε zero). It is the same Γ that enters x in §5.1, so the collapse plot uses no fitted parameter.

- **Code:** `gamma_point` study_mixed.py:88 and `gamma_job` study_tuning.py:489 bisect, in $\ln E$, for where (`eh_local`+`he_local`)/2 falls to half its $E = 0$ value.

5.3 Effective coupling, exponent, and the plateau criterion (definitions)

- $\Gamma_{\text{eff}} = \varepsilon_{\text{box}}/x(\det r)$, on two-Majorana points only (neutral, and relation §5.1 within 0.01): study_mixed.py:146-152.
  - Physics: if transport sees the box splitting, Γ_{eff} must stay smooth while ε varies by decades.
- $\Gamma \propto \tau^p$**:** $p = \ln(x_a/x_b)/\ln(\tau_b/\tau_a)$ at the same points of two maps, study_tuning.py:470-476.
  - Golden-rule expectation [**derived here**]: Nilsson's $\Gamma_i = 2\pi w_i^2$ with $w_i \propto \tau\,\psi_M(\text{contact})$ at fixed wavefunctions gives $p = 2$.
  - Measured: $p \approx 2.6$. Why it differs from 2 has **not** been established.
- **Plateau point [model choice]:**
  - $|\kappa - \frac{1}{2}| < 0.03$ and $|G| < 0.1\kappa$: study_tuning.py:394-396;
  - the stricter version, which also requires $Q = -1$ and $|\kappa - \frac{1}{2}(1 - \det r^2)| < 0.01$: study_mixed.py:266-269.
  - It marks an $\varepsilon \approx \Gamma$ region, never a protected plateau.

6. The cross-section: the infinitely long junction

6.1 Bloch Hamiltonian and gaps

[**derived here**] The junction is translation-invariant along x (Bloch k, units $1/a$). Across it there are n_S SC sites, the N strip and n_S SC sites, with hard walls ($W_S = 900$ nm per side; Scharf uses finite W_S too). The Bloch Hamiltonian is

$$H(k) = \bigoplus_j \left(h_j + V_{x,j}e^{ik} + V_{x,j}^\dagger e^{-ik}\right) + (\text{across hops}).$$

It is built from the same blocks as the transport code (tools.py:97-120).

- Topological gap (Scharf's Δ_{top}): $\min_k |E(k)|$.
- The gap dip is about $1/\xi_M$ wide, so the grid minimum is refined by a 1D minimisation: tools.py:187-220.

6.2 The invariant Q

[**source, verified**] Kitaev, Phys.-Usp. 44, 131 (2001), arXiv:cond-mat/0010440, **Eq. (26)**:

$$\mathcal{M}(H) = \operatorname{sgn}\operatorname{Pf}\tilde{B}(0)\,\operatorname{sgn}\operatorname{Pf}\tilde{B}(\pi),$$

where $\tilde{B}(q)$ is the real antisymmetric Majorana-basis Hamiltonian at the two particle–hole-invariant momenta. $\mathcal{M} = -1$ means unpaired Majoranas at the ends. Scharf, Sec. III, footnote: the quasi-1D junction is in class D and its invariant is the ground-state parity at $k = 0$, citing Kitaev.

[**derived here**] The Majorana basis for $\Psi = (c_\uparrow, c_\downarrow, c_\downarrow^\dagger, -c_\uparrow^\dagger)$ is the matrix `_W1` (tools.py:152-155):

- rows give $\gamma_1 = (c_\uparrow + c_\uparrow^\dagger)/\sqrt{2}$, $\gamma_2 = i(c_\uparrow - c_\uparrow^\dagger)/\sqrt{2}$, and the same for spin down;
- $M(k) = -iWH(k)W^\dagger$ is real antisymmetric at $k = 0, \pi$;
- a convention-dependent overall sign of Pf is the same at both momenta, so it cancels in Q.

Code

- Q: tools.py:177-184;
- Pfaffian sign by Parlett–Reid elimination with pivoting: tools.py:158-174. [**source, verified**] Wimmer, ACM TOMS 38, 30 (2012), arXiv:1102.3440, section "LTL^T decomposition of dense matrices using the Parlett-Reid algorithm".

Physics. Q belongs to the infinite junction: it does not know L or the probes. $\det r$ (§4.5) is the same invariant measured through a probe, and it tends to Q only for $L \gg \xi_M$.

7. Lengths and the splitting ε

7.1 Complex momenta: decay length ξ_M and oscillation k_{osc}

[derived here]

1. With nearest-neighbour hopping along the junction, $H(k) = H_0 + H_1 e^{ik} + H_1^\dagger e^{-ik}$.
2. A zero-energy solution $\psi_n = \lambda^n \varphi$ with $\lambda = e^{ik}$ (complex k) requires

$$(H_1\lambda^2 + H_0\lambda + H_1^\dagger)\varphi = 0.$$

3. With $\chi = \lambda\varphi$ this quadratic eigenproblem becomes the linear generalised one

$$\begin{pmatrix} 0 & 1 \\ -H_1^\dagger & -H_0 \end{pmatrix}\begin{pmatrix}\varphi \\ \chi\end{pmatrix} = \lambda \begin{pmatrix} 1 & 0 \\ 0 & H_1 \end{pmatrix}\begin{pmatrix}\varphi \\ \chi\end{pmatrix}.$$

4. Decaying modes have $|\lambda| < 1$. The slowest one sets the Majorana tail $|\psi| \propto e^{-x/\xi_M}$:

$$\xi_M = -\frac{a}{\ln|\lambda|}, \qquad k_{\text{osc}} = \frac{|\arg\lambda|}{a}.$$

The same mode construction underlies the mode-matching lead solvers that Lewenkopf–Mucciolo list ("eigenchannel decomposition or mode matching"). The code's lead fallback leads.py:123-135 uses it at finite E.

Getting H_0 and H_1 from three Bloch matrices [derived here].

- $H(0) = H_0 + H_1 + H_1^\dagger$, $H(\pi) = H_0 - H_1 - H_1^\dagger$ and $H(\pi/2) = H_0 + i(H_1 - H_1^\dagger)$.
- Hence $H_0 = \frac{1}{2}[H(0) + H(\pi)]$ and $H_1 = \frac{1}{2}\left[\frac{1}{2}(H(0) - H(\pi)) - i(H(\pi/2) - H_0)\right]$.

Code

- the block extraction: study_oscillation.py:61-66;
- the eigenproblem: :69-77;
- ξ_M, k_{osc}: :80-87;
- mixed SOC, 3 slowest modes: study_mixed.py:62.

Physics. The tail is a superposition of a few complex-k modes. The slowest one dominates at large distance; its $\mathrm{Re}\,k$ follows the outer Fermi point (CLAUDE.md, oscillation section).

7.2 The gap dip: ξ_M from the real band

[derived here]

1. Near the outer Fermi point k_0 the lowest band is a hyperbola,

$$E(k)^2 \simeq \Delta_{\text{out}}^2 + (\hbar v_A)^2 (k - k_0)^2, \qquad \hbar v_A = \sqrt{\Delta_{\text{out}} E''(k_0)}.$$

2. Continue it to $E = 0$: $k = k_0 \pm i\Delta_{\text{out}}/\hbar v_A$, so

$$\xi_{\text{hyp}} = \frac{\hbar v_A}{\Delta_{\text{out}}}.$$

3. This is the BCS form $\xi = \hbar v/\Delta$ with the Andreev-band velocity v_A, not v_F.
4. Beyond the hyperbola, E^2 is fitted by a degree-6 polynomial over $k_0 \pm 2/\xi_{\text{hyp}}$ and continued to its nearest complex zero (ξ_{poly}).

Code: study_oscillation.py:90-115.

Physics. ξ_M is a property of the narrow dip alone: a slow Andreev band (small v_A) plus a small gap at k_0. That is why $\hbar v_F/\Delta_{\text{top}}$ fails (CLAUDE.md).

7.3 The closed box: ε exactly

[**derived here**] A finite junction of $N = L/a$ slices with hard walls at both ends, and no probes:

$$H_{\text{box}} = \mathbb{1}_N \otimes H_0 + S \otimes H_1 + S^{\mathsf{T}} \otimes H_1^\dagger, \qquad S = \text{superdiagonal shift.}$$

ε is the smallest $|E|$, from sparse shift-invert at $\sigma = 0$. It is the splitting E_M of Nilsson's Eq. (1); the BdG levels come as $\pm E_M$.

- **Code:** study_oscillation.py:118-127.
- Cross-check: the lowest end-LDOS peak in the transport geometry, `splitting` tools.py:548-553, gives 4.0 μeV vs the box's 4.09 μeV (CLAUDE.md).

7.4 The overlap law $\varepsilon \approx A\Delta e^{-L/\xi_M} |\cos(k_{\text{osc}} L + \varphi_0)|$

[**source, verified**] Das Sarma, Sau & Stanescu, PRB 86, 220506(R) (2012), arXiv:1211.0539:

- **Eq. (1):** $\Delta E \sim \text{Re}[\Psi_l^\dagger(L/2)\hat{J}\Psi_r(L/2)]$.
- **Eq. (2):** $\Psi_l \propto e^{-x/\xi} e^{\pm i k_{F,\text{eff}} x}$.
- **Eq. (3):** $\Delta E \approx \hbar^2 k_{F,\text{eff}} \frac{e^{-2L/\xi}}{m\xi} \cos(k_{F,\text{eff}} L)$.

[**derived here**] The exponent depends on the convention. Here ξ_M is defined by $|\psi| \propto e^{-x/\xi_M}$ (§7.1), and the two ends are a distance L apart. The overlap of Eq. (1) then scales as e^{-L/ξ_M} and oscillates as $\cos(k_{\text{osc}} L + \varphi_0)$.

- The prefactor is not derived. A and φ_0 are fitted, two parameters for all fields: study_oscillation.py:254-263.
- The fit is a **test**: zeros spaced by π in $k_{\text{osc}} L$, a constant A.
- For the mixed SOC the phase holds but A drifts (CLAUDE.md, "THE PHASE HOLDS, THE AMPLITUDE DOES NOT").

7.5 End modes and Rayleigh–Ritz (study_mixed [amplitude])

[**derived here**]

1. **Left end mode.** $\psi_L(n) = \sum_{|\lambda_j|<1} c_j \varphi_j \lambda_j^{n+1}$, a zero-energy solution in the bulk by §7.1. The hard wall $\psi_L(-1) = 0$ means $\sum_j c_j \varphi_j = 0$, so c is the null vector of $[\varphi_j]$ (taken from the SVD).
2. **Right end mode.** The same with the growing modes: $\psi_R(n) = \sum_{|\lambda_j|>1} d_j \varphi_j \lambda_j^{n-N}$.
3. **Rayleigh–Ritz.** In $\text{span}\{\psi_L, \psi_R\}$ solve $H_{\text{RR}} u = E\, S_{\text{RR}} u$, which gives ε_{RR}.
4. **Boundary term.** $H\psi_L = 0$ inside, so truncating at N leaves only the missing neighbour:

$$\varepsilon \simeq \frac{|\psi_R(N-1)^\dagger H_1 \psi_L(N)|}{\|\psi_L\|\,\|\psi_R\|}.$$

5. **Semi-infinite norms** in closed form:

$$\sum_{m\geq 1} \Big| \sum_j c_j \varphi_j z_j^m \Big|^2 = \sum_{jk} c_j^* c_k (\varphi_j^\dagger \varphi_k) \frac{z_j^* z_k}{1 - z_j^* z_k}.$$

Code

- end modes: study_mixed.py:477-490;
- ψ_L, ψ_R: :493-507;
- norms: :510-516;
- RR and boundary term: :525.

Result: $\varepsilon_{\text{RR}}/\varepsilon_{\text{box}} = 0.9999$ wherever $L/\xi_M \geq 3$. The lobe heights are the overlap of the two end Majoranas.

7.6 LDOS and total DOS

- **[source, verified]** Gresta et al., **Eq. (8)**: LDOS $= \frac{1}{\pi}\mathrm{Im}\,\mathrm{Tr}\,\mathcal{G}^a$. That is equivalent to $-\frac{1}{\pi}\mathrm{Im}\,\mathrm{Tr}\,G^r$, which the code uses (spectra.py:149).
- Total DOS **[derived here]**:

$$\mathrm{Tr}\,G = \mathrm{Tr}\,g_0 + \mathrm{Tr}(K^{-1}W(g_0^2)_{PP}W^\dagger),$$

with K as in §2.2.

  - (g_0^2) is the off-diagonal block of $\begin{pmatrix} A & -1 \\ 0 & A \end{pmatrix}^{-1}$, which is still block-tridiagonal.
  - Code: tools.py:478-539.

8. Index: quantity → formula → code → source

quantity	formula	code	source
BdG model	Scharf Eq. (1), discretised	hamiltonians.py:25, ldos.py:196	Scharf 2019 Eq. (1)
E_Z, E_T	$g\mu_B B/2$, $\frac{\pi}{2}\hbar v_F/W$	ldos.py:148, :157	Scharf, Sec. VII text; Eq. (8)
Σ, Γ	$V_c^\dagger g_s V_c$, $i(\Sigma - \Sigma^\dagger)$	leads.py:239, rgf.py:293	Lewenkopf–Mucciolo Eqs. (16)–(17)
G^r (centre)	RGF plus ribbon Dyson step	rgf.py:115, :342	LM Sec. 3; Dyson step derived here
channels	$\mathrm{Tr}[\Gamma G^r \Gamma G^a]$ projected on e/h	rgf.py:373	Gresta Eq. (7)
G	$\mathcal{T}^{ee} - \mathcal{T}^{he}$	tools.py:290	Gresta Eqs. (3), (5)
κ	$\mathcal{T}^{ee} + \mathcal{T}^{he}$; kernel $\frac{3}{\pi^2}x^2 W_{\mathrm{el}}$	tools.py:290, transport.py:118	Gresta Eqs. (4), (6); Klees Sec. IV and App. C
local Andreev	$R_{eh} + R_{he}$	tools.py:290	derived; Law–Lee–Ng for $2e^2/h$
$\det r$	$-1 + i\Gamma^{1/2} G_{LL} \Gamma^{1/2}$	tools.py:266	LM Eqs. (5), (8); Fulga Table I
two-Majorana κ, $\det r$	$\frac{2x^2}{(1+x^2)^2}$, $\frac{x^2-1}{x^2+1}$	study_tuning.py:458	Nilsson Eqs. (1)–(6), $E=0$ derived here
Γ (direct)	HWHM of $R_{he}(E)$	study_mixed.py:88	derived from Nilsson Eqs. (4)–(6)
Q	$\mathrm{sgn}\,\mathrm{Pf}\tilde{B}(0)\,\mathrm{sgn}\,\mathrm{Pf}\tilde{B}(\pi)$	tools.py:177	Kitaev Eq. (26); Wimmer (Parlett–Reid)
ξ_M, k_{osc}	$-a/\ln\lvert\lambda\rvert$, $\lvert\arg\lambda\rvert/a$	study_oscillation.py:69	derived here
ξ_{hyp}	$\hbar v_A/\Delta_{\mathrm{out}}$	study_oscillation.py:90	derived here
ε	lowest $\lvert E\rvert$ of the closed box	study_oscillation.py:118	(= Nilsson's E_M)
overlap law	$A\Delta e^{-L/\xi_M}\lvert\cos(kL+\varphi_0)\rvert$	study_oscillation.py:254	Das Sarma et al. Eqs. (1)–(3); exponent convention derived here
$\varepsilon_{\mathrm{RR}}$	Rayleigh–Ritz in $\{\psi_L, \psi_R\}$	study_mixed.py:525	derived here
LDOS	$-\frac{1}{\pi}\mathrm{Im}\,\mathrm{Tr}\,G^r$	spectra.py:149	Gresta Eq. (8)

9. Sources

All papers were checked against their arXiv LaTeX source; equation numbers were checked against the printed PDF.

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99, 214503 (2019), arXiv:1904.08981. Used: Eq. (1), Eq. (8), $E_Z = g\mu_B B/2$ (Sec. VII), class D via Kitaev (Sec. III).

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3. R. L. Klees, D. Gresta, J. Sturm, L. W. Molenkamp et al., *Majorana-mediated thermoelectric transport in multiterminal junctions*, PRB 110, 064517 (2024), arXiv:2306.17845. Used: App. C (charge and heat currents), Sec. III (particle–hole symmetry of $\mathcal{T}$), Sec. IV (Onsager coefficients, $\mathcal{T}_{22}$).
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9. K. T. Law, P. A. Lee, T. K. Ng, *Majorana fermion induced resonant Andreev reflection*, PRL 103, 237001 (2009), arXiv:0907.1909. Used: $2e^2/h$.
10. A. Kitaev, *Unpaired Majorana fermions in quantum wires*, Phys.-Usp. 44, 131 (2001), arXiv:cond-mat/0010440. Used: Eq. (26).
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12. S. Das Sarma, J. D. Sau, T. D. Stanescu, *A Majorana smoking gun for the superconductor-semiconductor hybrid topological system* (arXiv title), PRB 86, 220506(R) (2012), arXiv:1211.0539. Used: Eqs. (1)–(3).
13. F. Pientka, A. Keselman, E. Berg, A. Yacoby, A. Stern, B. I. Halperin, *Topological superconductivity in a planar Josephson junction*, PRX 7, 021032 (2017), arXiv:1609.09482. The platform, as cited by Scharf.
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